

REVIEW

Are microplastics making people ill?

REFERENCES	STYLE	MADE WITH	VERIFICATION	COMPILED
6 verified	ELI5	Grounded v0.5.5	Crossref · claims	5 Sep 2026

SUMMARY

Tiny plastic pieces have been detected inside people, and experiments show that they can affect animals. Human studies link plastic in the body with some illnesses, but have not established how much everyday exposure contributes to disease.

1

Finding plastic inside the body

Studies have reported microplastics, tiny plastic pieces, in human tissues including lungs.¹ The detections leave open whether these pieces have harmed the body.¹ The unfinished line in Figure 1 marks that gap between detecting plastic and establishing harm.

Some studies did not report using checks for stray plastic entering samples during laboratory work, which makes those findings harder to interpret.¹ This uncertainty concerns the reliability of particular measurements; it does not mean every reported detection is false.¹

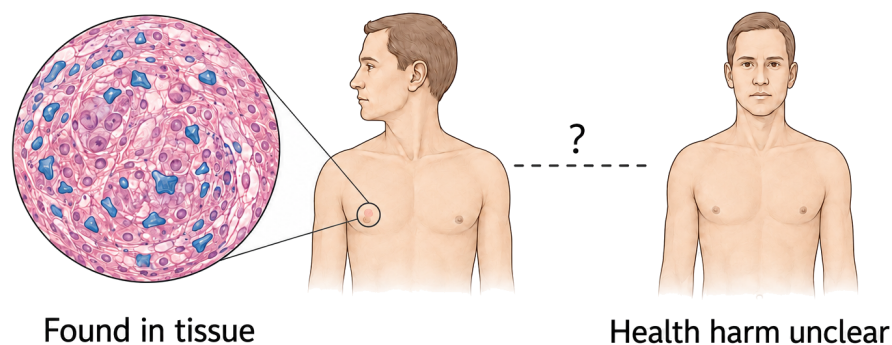


Figure 1. Finding plastic leaves its health effects unresolved. Human tissue detections leave causal health effects uncertain.^{1,2} The blue pieces represent detections, not measured sizes or amounts; the dashed line and question mark represent the unresolved causal connection. Neither torso represents a diagnosed patient.

2 Testing what the pieces can do

Experiments in mice have found changes in how their bodies handled sugar and fat, and in their ability to reproduce, after exposure to plastic particles.³ The doses and exposure times in those experiments may not closely match everyday human exposure, so the results do not establish human disease risk.³ Not every test found a clear effect: a rat study found no clear change in behaviour, although its small size and design left uncertainty.³

A review combining mostly animal evidence with a few human studies judged harm to digestion, breathing and reproduction to be “suspected”.⁴ That judgment identifies possible harm without establishing how much disease ordinary exposure causes. The same review could not judge effects on human birth outcomes because the evidence was too weak.⁴

3 Following people's health

Human follow-up studies connect a tissue measurement with what happens later. In a study, doctors tested material removed during surgery on narrowed blood vessels in the neck, then followed 257 patients for an average of about 34 months.⁵ A heart attack, stroke or death occurred in 30 of the 150 patients with detected plastic, compared with 8 of the 107 without detected plastic.⁵

The difference does not establish that plastic caused those events. The groups could also differ in other exposures, health conditions or behaviours that affected their later health.⁵ They were all patients undergoing surgery for artery disease, so this comparison cannot provide a risk estimate for the general population.⁵ The parallel paths in [Figure 2](#) show how the tissue tests defined the groups whose health was then followed.

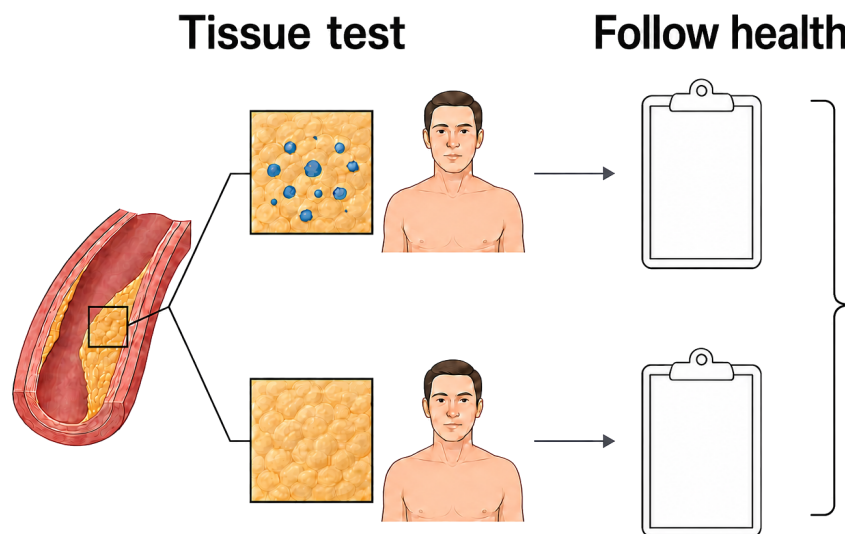


Figure 2. Tissue tests defined the groups being followed. The study followed patients with and without detected plastic in material removed during surgery.⁵ The two rows depict those groups, with arrows to later health records and a brace marking their comparison. The drawing encodes neither patient numbers nor event counts. Other differences between the groups can affect the comparison.^{5,2}

4 What remains uncertain

The artery study also could not firmly rule out stray plastic entering samples in the laboratory.⁵ Across human studies, uneven measurements and unaccounted-for differences between people make it difficult to separate plastic's possible effects from other causes of illness.² The findings also vary by outcome: some studies of nasal conditions found no clear relationship between more measured plastic and worse symptoms.² Small studies and inadequate exposure measurements still prevent a reliable estimate of health risk.⁶

Experiments and human associations support concern about possible harm. How much everyday exposure contributes to human illness remains uncertain.

Scope and methods: This short narrative review uses the retained corpus from PubMed searches on 4 September 2026, covering human studies, animal experiments, reviews and contrary findings. Five cited papers were read in full; one was available only as an abstract. Animal results were not used to calculate human risk. This editorial revision includes no new literature search.



MADE WITH GROUNDED

You can make one of these too. Grounded is a free skill — a set of instructions you give to an AI helper so it knows how to do a job properly. This one makes it go and find real science papers, read them, and check that every single one is real before it writes anything.
github.com/jostelzer/grounded

References

6 · VERIFIED VIA CROSSREF

1. **Roslan NS, Lee YY, Ibrahim YS, Tuan Anuar S, Yusof KMKK, Lai LA, Brentnall T (2024)** Detection of microplastics in human tissues and organs: A scoping review. *Journal of Global Health*. <https://doi.org/10.7189/jogh.14.04179> · 5 claims · full text
2. **Tran HAA, Ng JJ, Tang DH, Yong CT, Ho AFW (2026)** Health impacts of micro- and nanoplastics in humans: systematic review of in vivo evidence. *Environmental Health*. <https://doi.org/10.1186/s12940-026-01282-y> · 4 claims · full text
3. **Liu W, Zhang B, Yao Q, Feng X, Shen T, Guo P, Wang P, Bai Y, Li B, Wang P, Li R, Qu Z, Liu N (2023)** Toxicological effects of micro/nanoplastics on mouse/rat models: a systematic review and meta-analysis. *Frontiers in Public Health*. <https://doi.org/10.3389/fpubh.2023.1103289> · 3 claims · full text
4. **Chartres N, Cooper CB, Bland G, Pelch KE, Gandhi SA, BakenRa A, Woodruff TJ (2024)** Effects of Microplastic Exposure on Human Digestive, Reproductive, and Respiratory Health: A Rapid Systematic Review. *Environmental Science & Technology*. <https://doi.org/10.1021/acs.est.3c09524> · 2 claims · full text
5. **Marfella R, Prattichizzo F, Sardu C, Fulgenzi G, Graciotti L, Spadoni T, D'Onofrio N, Scisciola L, La Grotta R, Frigé C, Pellegrini V, Municinò M, Siniscalchi M, Spinetti F, Vigliotti G, Vecchione C, Carrizzo A, Accarino G, Squillante A, Spaziano G, Mirra D, Esposito R, Altieri S, Falco G, Fenti A, Galoppo S, Canzano S, Sasso FC, Matacchione G, Olivieri F, Ferraraccio F, Panarese I, Paolisso P, Barbato E, Lubritto C, Balestrieri ML, Mauro C, Caballero AE, Rajagopalan S, Ceriello A, D'Agostino B, Iovino P, Paolisso G (2024)** Microplastics and Nanoplastics in Atheromas and Cardiovascular Events. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2309822> · 7 claims · full text
6. **Lamoree MH, van Boxel J, Nardella F, Houthuijs KJ, Brandsma SH, Béen F, van Duursen MBM (2025)** Health impacts of microplastic and nanoplastic exposure. *Nature Medicine*. <https://doi.org/10.1038/s41591-025-03902-5> · 1 claim · abstract ■

6 SOURCES · EVERY DOI RESOLVED · RETRACTION-SCREENED VIA CROSSREF

20 CITED SENTENCES · 22 SOURCE CHECKS · 21 SUPPORTED AT FULL TEXT · 1 AT ABSTRACT · 0 PARTIAL · 0 CONTRADICTED

GROUNDED V0.5.5 · COMPILED 5 SEPTEMBER 2026